

Development of an Artificial Intelligence (AI) Action Plan

Response to Request for Information (RFI)

Submitted to:

National Science Foundation (NSF)
Office of Science and Technology Policy (OSTP)
Networking and Information Technology Research and Development (NITRD)
National Coordination Office (NCO)
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1.0 INTRODUCTION

KBR offers the following response to the Networking and Information Technology Research and Development (NITRD) National Coordination Office (NCO), National Science Foundation's (NSF) Request for Information (RFI), on behalf of the White House Office of Science and Technology Programs (OSTP). NITRD is seeking input on the highest priority policy actions that should be in the AI Action Plan directed by a Presidential Executive Order on January 25, 2025. The Order directs the OSTP to develop a plan that defines priority policy actions needed to sustain and enhance America's AI dominance and leadership to promote human flourishing, economic competitiveness, and national security. The Order further revoked a previous Executive Order which hampered the private sector's ability to innovate in AI by imposing burdensome government requirements restricting private sector AI development and deployment.

With decades of increasingly AI-enabled, global, mission critical logistics and infrastructure operations, construction, defense and intelligence, and science support services KBR is positioned to provide practical recommendations to enhance OSTP AI Action plan policy development. KBR appreciates OSTP's efforts to engage businesses like ours in the formulation of the plan. It is imperative that businesses in key industries continue to be consulted as the planning effort continues.

As a global superpower, America's traditional role has been to leverage economic and military dominance to ensure our way of life. This included a hefty investment in technologies that propelled our societal advancements. AI tools have advanced significantly over the last several years and potential adversaries are leveraging these tools as a method to even the playing field with an eye to surpassing our technical superiority. U.S. regulatory burdens make it more difficult to mitigate this existential threat. This should give the Nation significant pause, and NSF should seize the opportunity to identify clear objectives and policy initiatives per the Executive Order (EO), including.

2.0 RECOMMENDATIONS

In response to this RFI, KBR offers three recommendations for consideration in developing the AI Plan:

1. Prioritize AI deployment in national defense applications,
2. Leverage established NSF framework and programs to advance AI leadership, and
3. Address energy and other infrastructure needs to enhance the U.S. domestic AI industry.

RECOMMENDATION 1—PRIORITIZE AI DEPLOYMENT FOR NATIONAL DEFENSE APPLICATIONS

KBR's viewpoint is that the AI Action Plan should place high priority on AI support to national defense needs, including immediate access to AI-enabled critical information systems, AI-enabled decision intelligence platforms, and AI-enabled secure architecture. Ultimately, the AI Action Plan needs to address how we create an environment in which American ingenuity will allow us to stay ahead of our potential adversaries and retain our leadership position to keep our homeland and our allies safe and secure.

KBR, our Warfighters, and global allies need AI-driven decision support systems that enable our ability to operate in a wide variety of environments. To ensure America's dominance, the AI Action Plan should enable continued development of these AI attributes for the following areas:

1. Develop AI-driven decision support systems to:

- a. Connect the developed world with remote and austere environments enabling rapid transfer of critical data over limited communications architecture

- b. Quickly gain insights into changing conditions impacting the mission which may not be readily understood by those on the ground
- c. Take historical and current information to predict how conditions could impact the mission in the future
- d. Optimize these mission critical courses of action across an infinite number of tangible and intangible factors and quickly generate recommended courses of action with the highest probabilities of success
- e. Execute these decision support systems cycle faster than our competition
- f. Ensure security and fidelity of the decision support system.

2. *Develop AI-powered autonomous systems for maintaining infrastructure in austere and remote environments to:*

- a. Assess the current infrastructure condition and requirements and identify specific maintenance tasks that can be automated using AI-powered systems
- b. Collaborate with AI and robotics experts to design autonomous systems tailored to the remote critical mission requirements to ensure these systems are capable of handling various maintenance tasks, such as equipment inspection, repairs, and environmental monitoring
- c. Develop prototypes of the autonomous systems and conduct rigorous testing in controlled environments allows for gathering feedback from researchers and technicians to refine the systems
- d. Gradually deploy autonomous systems across global mission critical locations and implementing monitoring mechanisms to track performance and identify areas for improvement is crucial
- e. Provide training and support for research, Warfighter and contractor station staff on how to operate and maintain the autonomous systems, as well as establishing a support system for troubleshooting and ongoing maintenance, ensures smooth operation.

3. *Develop AI-driven logistics tools to:*

- a. Establish predictive maintenance systems to ensure continuous operation by identifying critical equipment and components that require regular maintenance and prioritizing them based on their impact on overall operations and potential failure risks.
- b. Deploy IoT sensors on critical equipment to collect real-time data on performance and condition, such as vibration, temperature, and pressure
- c. Develop AI algorithms to analyze sensor data and predict potential failures using historical data and machine learning techniques improves prediction accuracy.
- d. Integrate predictive maintenance algorithms with maintenance scheduling systems and automating maintenance alerts and scheduling based on predictive insights ensures timely and effective maintenance.
- e. Leverage AI-enabled optimization techniques to provide the best and fastest mission critical decision support processes to ensure the right product/part is delivered at the right place at the right time in the right quantities.
- f. Provide training for maintenance teams using predictive maintenance tools and interpreting AI-generated insights equips them to perform their tasks efficiently.

- g. Continuously monitor the performance of the predictive maintenance systems and collect feedback from maintenance teams to facilitate improvements to the algorithms and processes.

4. *Develop AI-enabled remote monitoring climate and environmental monitoring capabilities to:*

- a. Identify key monitoring parameters, such as temperature, humidity, air quality, and soil moisture, in consultation with environmental scientists and researchers to ensure comprehensive coverage
- b. Select appropriate sensors and IoT devices for data collection and deploy them strategically across different locations to ensure accurate and representative data. Computer vision augments the collection of real time data, allowing less human labor to identify critical environmental changes
- c. Develop AI algorithms to analyze the collected data and generate actionable insights, including detecting anomalies, predicting trends, and providing real-time alerts, is essential
- d. Integrate the AI-driven monitoring networks with existing data management and reporting systems to ensure seamless data flow and accessibility for researchers and decision-makers
- e. Regularly review the performance of the monitoring networks, make necessary adjustments and update AI algorithms and sensor configurations based on new research and technological advancements.

With these objectives in mind, the right AI policies can help achieve success in various remote and austere mission environments such as expeditionary military endeavors. Additionally, the right policies will spur economic acceleration of dual use technology in the civilian sectors including intelligent transportation, utility grids and infrastructure, robotics, human-AI interaction, and various use cases in neuroscience, biology, chemistry, physics, manufacturing, and agriculture.

RECOMMENDATION 2—LEVERAGE ESTABLISHED NSF FRAMEWORK AND PROGRAMS TO ADVANCE AI LEADERSHIP

NSF has been in a key leadership role in supporting fundamental and translational research in AI technologies, developing AI systems to enhance learning for all and grow the next generation of talent to advance the U.S. AI R&D workforce, and providing access to advanced cyberinfrastructure including scalable computing resources and high-quality and accurate training datasets to advance AI research and education. NSF programs, such as the NSF's United States Antarctic Program (USAP), that conduct important research in remote and harsh environments can greatly benefit from AI advancements to continue deriving the maximum scientific benefit while significantly reducing costs and improving data transparency and reliability.

Through years of investment and national technology development leadership, NSF established a framework to ensure America's continued innovation in AI methods, application of AI techniques and tools, democratizing AI research resources, trustworthy and ethical AI, education and workforce development, and partnerships to accelerate progress. NSF has held a leadership role in supporting and funding AI research which is helping to transform every industry and function. Recent NSF investments included several accelerators including:

- National AI R&D Strategic Plan,
- National AI Initiative Act of 2020,
- National AI Research Resource (NAIRR) Task Force Pilot enhancing access to computational resources, software, datasets, models, services and training to advance AI research and the use of AI in research and education,

- Twenty-five AI institutes covering over 500 funded collaborative institutions across the US and around the world.
- Collaboration with NIST's leadership to develop the framework and security protocols necessary for the safe and secure development of AI.

KBR recommends building on the established NSF AI framework which focuses on the following:

- **Data and Data Centers**—Establishment of the most comprehensive network of data sets through open and secure data initiatives.
- **Decision Support Intelligence and Computational Hardware**—Focus on AI-enhanced digital Model Based System Engineering (MBSE) systems to create digital models that replicate life situations. Ensure the fastest, most comprehensive AI-enhanced data calculations based on ultra-high speed hardware processing and optimization strategies. Leverage annual competition with large and small industry actors, universities, and other actor collaborations to regularly capture best-in-class algorithms and toolsets.
- **Speed of Information Flow**—Enhance global communication speed through AI-enabled communication pathway optimization. Focus on AI-enabled data prioritization, data compression, most efficient route selection, reduced energy power levels required and energy consumption.
- **Predictive Analytics**—Reinforce development of AI-enabled 'over-the-horizon' impacts to localized operations, allowing time and space to mitigate potential impacts to organizational production. Improve logistics operations through AI-enabled early identification of supply and materials needs, enhanced intelligent transportation methods and operations, most efficient route selection strategies.
- **Cybersecurity**—Enhance AI-enabled cyber threat identification, containment, mitigation, and elimination systems. Ensure trustworthy and ethical AI architecture and systems.

KBR believes that continued funding of these and other NSF AI-related programs should continue as an integral part of maintaining America's AI leadership position.

RECOMMENDATION 3—ADDRESS ENERGY AND OTHER INFRASTRUCTURE NEEDS TO ENHANCE THE U.S. DOMESTIC AI INDUSTRY

The increasing energy demands of AI workloads necessitate a multi-faceted strategy to improve energy efficiency and sustainability. This strategy includes deploying low-power AI solutions, implementing microgrids, and optimizing energy usage in AI infrastructure. Alternative power sources such as nuclear, renewables, and natural gas turbines will be a vital part of powering the ongoing AI boom. It is crucial that the regulatory and licensing/permitting structure around these energy sources is streamlined so that sufficient power is available where it is needed and on the timeframe that is required to maintain U.S. competitiveness.

By utilizing enhanced hardware components, algorithmic optimization, and software efficiency enhancements, the energy cost of operating an AI system can be significantly reduced. Application-Specific Integrated Circuits (ASICs), which are custom-designed chips optimized for AI workloads with lower power consumption, play a crucial role in this strategy. ASICs are widely used in cryptocurrency mining, AI acceleration, 5G telecommunications, autonomous vehicles, and advanced signal processing applications. In AI, they serve as dedicated accelerators for neural networks, reducing computation time and energy consumption compared to traditional processors. Implementing AI-driven Electronic Design Automation (EDA) tools can improve chip layout, transistor density, and power efficiency, with AI optimizing chip layouts in hours instead of weeks, significantly reducing production bottlenecks. High-performance computing (HPC) clusters can

be used to speed up ASIC simulation and validation, allowing for multiple design variations to be tested simultaneously and reducing development time. Transitioning to Extreme Ultraviolet (EUV) lithography reduces the complexity of photolithography steps, improving precision and yield rates. Multi-patterning techniques allow smaller transistors to be printed on wafers, increasing chip density per wafer.

3.0 CONCLUSION

KBR supports the initiative to develop a national strategy for ensuring U.S. leadership in the AI arena for the betterment of all Americans, continued economic competitiveness, and national defense. Ongoing transparency as planning continues is essential to the ultimate success and buy-in of the national strategy and KBR looks forward to continuing to monitor its progress and providing additional information and perspectives into this process.